

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1443

Baytown Energy Center 100 kV
March 25, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1443. The Interconnection Customer proposes a 35.5 MW battery energy storage facility interconnecting at the Baytown Energy Center 100 kV substation in the Illinois area. The request was received on April 03, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 548 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$710,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: San Leandro U - Rockport Sta 100 KV
Contingency: Hockinson - Basalt 100 KV L
Loading: 108.3% (Rating: 133 MVA)
DFAX: 6.3%
Pre-project loading: 106.6%
2. Facility: Basalt - Hockinson 100 KV
Contingency: San Leandro U - Rockport Sta 100 KV Line
Loading: 104.5% (Rating: 132 MVA)
DFAX: 6.1%
Pre-project loading: 102.9%
3. Facility: Rockport Station - Midlothian South 100 KV L
Contingency: Basalt - Hockinson 100 KV
Loading: 104.4% (Rating: 128 MVA)
DFAX: 6.3%
Pre-project loading: 102.7%
4. Facility: Easy Road - Phelps Dodge 230 KV
Contingency: Phelps Dodge - Nrg Rockford I 230 KV
Loading: 103.0% (Rating: 625 MVA)
DFAX: 13.3%
Pre-project loading: 102.3%
5. Facility: Midlothian South - Foreman N 100 KV
Contingency: Basalt - Hockinson 100 KV L
Loading: 102.0% (Rating: 123 MVA)
DFAX: 6.3%
Pre-project loading: 100.2%

6. Facility: Talenenergy Martins Crk Llc Harrisburg 500 kV/345 kV TX
 Contingency: Westwood Generation Llc - Coronet 500 KV
 Loading: 101.9% (Rating: 1,826 MVA)
 DFAX: 5.6%
 Pre-project loading: 101.8%
7. Facility: Basalt - Ross Avenue 100 KV
 Contingency: San Leandro U - Rockport Sta 100 KV
 Loading: 101.1% (Rating: 128 MVA)
 DFAX: 6.1%
 Pre-project loading: 99.4%
8. Facility: Rio Hondo - Foreman North 100 KV Line
 Contingency: Basalt - Hocknson Substation 100 KV
 Loading: 100.4% (Rating: 115 MVA)
 DFAX: 6.3%
 Pre-project loading: 98.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rockport Station - San Leandro U 100 KV	Reconductor line	\$3,020,000	2% (\$72,000)
Basalt - Hockinson 100 KV	Reconductor line	\$1,799,000	2% (\$35,000)
Rockport Station - Midlothian S 100 KV	Reconductor line	\$1,961,000	2% (\$47,000)
Phelps Dodge Substation - Easy Road 230 KV L	Reconductor line	\$12,946,000	2% (\$307,000)
Midlothian South - Foreman N 100 KV	Reconductor line	\$2,159,000	2% (\$51,000)
Talenenergy Martins Crk Llc Harrisburg 500 kV/345 kV TX	Replace transformer with higher-rated unit	\$44,543,000	0% (\$142,000)
Basalt - Ross Avenue 100 KV line	Reconductor line	\$1,674,000	2% (\$33,000)
Rio Hondo - Foreman North 100 KV L	Reconductor line	\$962,000	2% (\$23,000)
	Total allocated to project		\$710,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.